

Underwater Vision-Based Surveillance Using ML and AI

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Day 22 — Final Model / Pipeline Freeze

1. Objective

Freeze the final project methodology and selected model configurations, based on validation evidence and robustness.

The freeze covers:

- Final model configurations
- Preprocessing methodology
- Dataset split strategy
- Evaluation methodology
- Training configurations
- Technical selection rationale
- Final trained model checkpoints
- Final project limitations
- Reproducibility information

2. Final Project Workflow

Raw Dataset → Quality Check → Annotation Review → Existing Split Preservation → Resize (640×640) → RGB Conversion → Pixel Normalization (0–1) → Selected Image Enhancement → YOLOv8n Detection → Evaluation → Final Analysis

3. Final Dataset Information

Dataset	Classes	Train	Validation	Test
DIATAquarium	11	8,121	773	Not available locally
Aquatic Plant	1	1,892	179	89
Well	4	3,795	191	184

Dataset Strategy

- All three datasets are used separately — different class structures and characteristics
- Supplied dataset splits are preserved; no new random split introduced after experiments
- Train = model learning; Validation = configuration selection; Test = final evaluation only, where available
- Validation and test data are never used for training

DIATAquarium Test-Set Limitation

The local DIATAquarium copy does not contain a separate test folder. The final DIAT configuration is therefore reported using the available validation set. No separate held-out DIAT test result is claimed.

4. Frozen Preprocessing

Component	Final Setting
Image Size	640×640
Color Format	RGB
Pixel Normalization	0–1
Raw Dataset	Preserved unchanged
Training Augmentation	Applied only during training
Validation Augmentation	Not applied
Test Augmentation	Not applied

Training Augmentation

- Horizontal flip
- Small rotation
- Scale variation
- Brightness / contrast variation

Raw datasets remain unchanged; preprocessing is applied only to working copies.

5. Image Enhancement

Classical enhancement method: White Balance + CLAHE.

Original Image → White Balance → CLAHE → Enhanced Image → YOLOv8n

Dataset-Specific Enhancement

- DIATAquarium: Original Image → YOLOv8n (no enhancement)
- Aquatic Plant: Original Image → WB+CLAHE → YOLOv8n
- Well: Original Image → WB+CLAHE → YOLOv8n

Enhancement was applied only where validation evidence supported it — it was not forced on every dataset.

6. Final Model Selection Criteria

Primary selection metric: Validation mAP@0.5:0.95. Supporting metrics: Precision, Recall, mAP@0.5.

Selection was based on three criteria:

- Validation evidence — the configuration with the highest validation mAP@0.5:0.95 was preferred for each dataset

- Robustness — no severe divergence between training and validation performance was observed for the selected configurations (see Day 18 overfitting discussion); deeper training-curve analysis is recommended for future work
- Appropriate simplicity — YOLOv8n is retained as the single detector architecture across all three datasets; enhancement is applied only where validation evidence supported it (Aquatic Plant, Well), avoiding unnecessary pipeline complexity for DIAT

7. Final Configuration — DIATAquarium

Configuration: Original Baseline

Original Image → YOLOv8n

Parameter	Value
Detector	YOLOv8n
Enhancement	None
Image Size	640×640
Batch Size	16
Epochs	30
Optimizer	Auto-selected by Ultralytics (AdamW)
Learning Rate	Auto-selected, approximately 0.000667

Metric	Validation Result
Precision	90.02%
Recall	92.09%
mAP@0.5	92.72%
mAP@0.5:0.95	65.90%

Selection reason: the original YOLOv8n configuration was retained for DIAT because the WB+CLAHE configuration reduced validation performance across all four metrics. Final DIAT configuration: Original Image → YOLOv8n.

8. Final Configuration — Aquatic Plant

Configuration: H03

Original Image → WB+CLAHE → YOLOv8n

Parameter	Value
Detector	YOLOv8n
Enhancement	WB+CLAHE
Learning Rate	0.0005
Batch Size	32
Epochs	30

Parameter	Value
Image Size	640×640
Optimizer	AdamW

Metric	Validation Result
Precision	95.4%
Recall	96.1%
mAP@0.5	98.5%
mAP@0.5:0.95	71.5%

Selection reason: H03 achieved the highest validation mAP@0.5:0.95 among the Aquatic Plant tuning experiments (H01–H03).

9. Final Configuration — Well

Configuration: H04

Original Image → WB+CLAHE → YOLOv8n

Parameter	Value
Detector	YOLOv8n
Enhancement	WB+CLAHE
Learning Rate	0.001
Batch Size	16
Epochs	30
Image Size	640×640
Optimizer	AdamW

Metric	Validation Result
Precision	35.7%
Recall	40.1%
mAP@0.5	36.8%
mAP@0.5:0.95	16.8%

Selection reason: H04 achieved the highest validation mAP@0.5:0.95 among the tested Well configurations (H04–H06).

10. Final Configuration Summary

Dataset	Config	Enh.	LR	Batch	Epochs	Prec.	Recall	mAP50	mAP50-95
Aquatic Plant	H03	WB+CLAHE	0.0005	32	30	95.4%	96.1%	98.5%	71.5%
Well	H04	WB+CLAHE	0.001	16	30	35.7%	40.1%	36.8%	16.8%
DIATAquarium	Baseline	None	—	16	30	90.02%	92.09%	92.72%	65.90%

11. Final Model Checkpoints & Saved Artefacts

The trained final model checkpoints, configuration record, and freeze summary were saved and verified on Google Drive.

Artefact	Path	Size
DIATAquarium checkpoint	/Day22_Final_Model/DIAT_Baseline/weights/best.pt	~5.95 MB
Aquatic Plant H03 checkpoint	/Day22_Final_Model/H03_Training/Aquatic_Plant_H03/weights/best.pt	~5.96 MB
Well H04 checkpoint	/Day22_Final_Model/H04_Training/Well_H04/weights/best.pt	~5.96 MB
Final configuration CSV	/Day22_Final_Model/final_configuration_summary.csv	—
Final freeze summary	/Day22_Final_Model/Day22_Freeze_Summary.md	—

Final Artifact Verification

Artefact	Status
Final configuration CSV	Verified
Final freeze summary	Verified
DIAT YOLOv8n checkpoint	Verified
Aquatic Plant H03 checkpoint	Verified
Well H04 checkpoint	Verified

12. Evaluation Metrics Used

- Precision — proportion of predicted detections that are correct
- Recall — proportion of actual objects successfully detected
- mAP@0.5 — Mean Average Precision at IoU threshold 0.5
- mAP@0.5:0.95 — Mean Average Precision averaged across IoU thresholds 0.5–0.95 (primary selection metric)

- Inference Speed (ms/image, FPS) — measured to understand computational performance

13. Enhancement Analysis — Original vs. WB+CLAHE

Dataset	Condition	Precision	Recall	mAP@0.5	mAP@0.5:0.95
DIAT	Original	90.02%	92.09%	92.72%	65.90%
DIAT	WB+CLAHE	87.70%	90.20%	91.10%	64.00%
Aquatic Plant	Original	94.64%	93.52%	98.05%	70.79%
Aquatic Plant	WB+CLAHE	94.60%	97.90%	98.70%	70.70%
Well	Original	35.60%	36.80%	36.50%	16.40%
Well	WB+CLAHE	36.30%	37.70%	35.20%	15.70%

Main finding: the effect of WB+CLAHE is dataset-dependent. It improved some detection measures for Aquatic Plant and Well, but reduced overall validation mAP for DIAT. Image enhancement should therefore be evaluated together with the downstream detection task, rather than assuming improved visual quality automatically improves detection.

14. Final Project Limitations

- The local DIATAquarium copy does not contain a separate test set
- Source-level overlap exists within the supplied dataset splits
- The Well dataset has strong class imbalance
- Fish and Fishes in DIAT are visually similar and may cause class confusion
- WB+CLAHE does not consistently improve detection across all datasets
- The individual effects of White Balance and CLAHE were not isolated
- Water-Net and FUNIE-GAN were studied as candidate methods but not implemented in the final experiments
- Video data is currently unavailable — object tracking is outside the current project scope
- The final project focuses on image-based object detection using YOLOv8n
- PSNR, SSIM and UIQM were considered for enhancement evaluation, but no quantitative PSNR/SSIM/UIQM results are claimed for the supplied project datasets

15. Final Scope

Underwater Image Enhancement + Object Detection, using:

- Three supplied underwater datasets
- WB+CLAHE for selected enhanced pipelines
- YOLOv8n for object detection
- Validation-based model selection
- Precision, Recall, mAP@0.5, mAP@0.5:0.95 for detection evaluation

Tracking is not included because video data is currently unavailable.

16. Reproducibility

The following information has been preserved:

- Dataset structure and split strategy
- Preprocessing configuration
- Enhancement method
- YOLOv8n model configuration (learning rate, batch size, epochs, image size)
- Validation metrics
- Final model checkpoint paths
- Final configuration CSV and freeze summary
- Documented limitations

Raw datasets remain unchanged. Final model checkpoints are stored separately from the raw datasets.

17. Final Frozen Pipeline

DIATAquarium

Original Image → 640×640 Preprocessing → YOLOv8n → Detection Evaluation

Aquatic Plant

Original Image → 640×640 Preprocessing → White Balance → CLAHE → YOLOv8n → Detection Evaluation

Well

Original Image → 640×640 Preprocessing → White Balance → CLAHE → YOLOv8n → Detection Evaluation

18. Final Conclusion

The final model and pipeline configurations were selected using validation evidence, robustness, and appropriate simplicity, with mAP@0.5:0.95 as the primary metric.

- DIATAquarium: Original Image + YOLOv8n
- Aquatic Plant: WB+CLAHE + YOLOv8n (H03)
- Well: WB+CLAHE + YOLOv8n (H04)

The final trained model checkpoints have been saved and verified. The preprocessing, dataset strategy, evaluation protocol, and model configurations are now frozen for final evaluation and project reporting.

DAY 22 FINAL VERIFICATION: PASSED

[12]

✓ Os



```
import os
paths = {
    "DIAT Baseline": "/content/drive/MyDrive/Day22_Final_Model/DIAT_Baseline/weights/best.pt",
    "Aquatic Plant H03": "/content/drive/MyDrive/Day22_Final_Model/H03_Training/Aquatic_Plant_H03/weights/best.pt",
    "Well H04": "/content/drive/MyDrive/Day22_Final_Model/H04_Training/Well_H04/weights/best.pt",
}
all_ok = True
for name, p in paths.items():
    exists = os.path.exists(p)
    size = os.path.getsize(p)/(1024*1024) if exists else 0
    print(f"{name}: {'FOUND' if exists else 'MISSING'} ({size:.2f} MB)")
    all_ok = all_ok and exists
print("\nDAY 22 FINAL VERIFICATION:", "PASSED" if all_ok else "FAILED")
```

▼

```
*** DIAT Baseline: FOUND (5.95 MB)
    Aquatic Plant H03: FOUND (5.96 MB)
    Well H04: FOUND (5.96 MB)
```

DAY 22 FINAL VERIFICATION: PASSED